Parsa Esmati

FINAL-YEAR PHD IN ENGINEERING (MACHINE LEARNING) · UNIVERSITY OF BRISTOL
Machine Learning and Computer Vision Group
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Education

Sep 2022 – Nov 2026 **University of Bristol, Bristol** · PhD in Engineering (Machine Learning)
Thesis: *Representation Geometry and Dynamics for Efficient Latent Generation*
Advisor: Prof. Majid Mirmehdi

Sep 2018 – Jun 2022 **University of Bristol, Bristol** · BEng and MEng in Engineering
Control theory, dynamics, robotics and computational methods

Professional Experience

Jun 2025 – Oct 2025 **Microsoft Research, Cambridge** · Research Intern
Pretraining the video generative world model for multi-view, multi-agent interactions on 100+ GPUs.
Re-engineering of the data pipeline to remove redundant caching, cutting data-loading stalls, which led to enhanced GPU utilisation during multi-node pretraining.
Research into tokenizer reconstruction and discrete generation (under review, ICLR 2027).

Sep 2022 – present **University of Bristol, Bristol** · Doctoral Researcher and Teaching Assistant
Research into representation learning and generation difficulty.

Publications

The Invisible Hand of Physics: When Video Diffusion Models Know More Than They Show
Parsa Esmati, Sourav Nath, Katja Hofmann, Derek Nowrouzezahrai, Samira Ebrahimi Kahou, Majid Mirmehdi
NeurIPS 2026, **Spotlight (top 3%)** · arXiv

Probability-Conserving Flow Guidance
Parsa Esmati, Junha Hyung, Amirhossein Dadashzadeh, Jaegul Choo, Majid Mirmehdi
NeurIPS 2026 · arXiv

SEA: State-Exchange Attention for High-Fidelity Physics Based Transformers
Parsa Esmati, Amirhossein Dadashzadeh, Vahid Goodarzi Ardakani, Nicolas Larrosa, Nicolò Grilli
NeurIPS 2024

Co-STAR: Collaborative Curriculum Self-Training with Adaptive Regularization for Source-Free Video Domain Adaptation
Amirhossein Dadashzadeh, Parsa Esmati, Majid Mirmehdi
WACV 2025 · arXiv

Visual Tokenization with Latent Anchored Binary Quantization
Parsa Esmati, Junha Hyung, Emanuele Aiello, Raluca Stevenson, Tao He, Jiang Bian, Majid Mirmehdi, Katja Hofmann
Under review, **ICLR 2027** · OpenReview

HuPo: Reconstructing Human Body via Sensor-Agnostic Point Cloud Restoration
Qiushuo Cheng, Zhihao Sun, Parsa Esmati, Majid Mirmehdi
Under review, **CVPR 2027**

The Nesting Hypothesis: How Latent Geometry Shapes Generative Difficulty
Parsa Esmati, Majid Mirmehdi
Preprint

Awards and Service

Isambard-AI Phase 2	Competitive compute allocation on the UK national AI supercomputer.
Reviewer	NeurIPS, ICLR, ICML.

Engineering Stack

Languages	Python, C++, C, MATLAB.
Frameworks	PyTorch, DeepSpeed, PyTorch Lightning, JAX; NumPy, Pandas, scikit-learn.
Models	Transformer generative models (DiT video world models, ViT tokenizers), diffusion and flow models; fine-tuning and sampling-time control of SD3, Flux and vision-language models.
Training at scale	Multi-node training with DDP, FSDP and DeepSpeed Ulysses sequence parallelism; SLURM.
Compute	Microsoft Azure; Isambard-AI.
DevOps	Git and GitHub, CI/CD, Docker, Linux.